

PSAT coding sheet 16/3/26

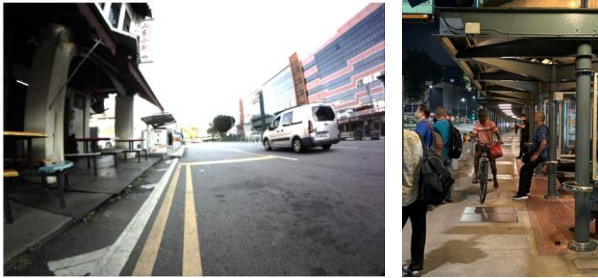
- 1)** Videos and photos should be recorded during off peak to prevent obscuring views.
- 2)** It is still useful to code as 10m interval as there could be observations like potholes, obstacles, unless it is along rural roads.
Any bigger interval might miss important features like driveway.
- 3)** If there are more than 1 coder, coders have to discuss on their coding regularly to iron out discrepancies.
- 4)** For each photo, code the most prominent type (path or junction) or look at the photos in front or back to see if junction has already been coded; if no, then code this photo as junction.

Attribute	Code	Coding guidance	Local guidance
Area Type Urban Suburban Rural Industrial Recreational	1 2 3 4 5	Select the predominant area type to assist with analysis. Standard options may be altered to better meet the local context. Do not change area type over short lengths (less than 100m) Urban • High-density residential or commercial development • High street connectivity, frequent intersections • Typically within city centres or dense neighbourhoods • OSM <code>landuse=residential</code>, <code>commercial</code>, <code>retail</code> Suburban • Lower density residential areas with moderate connectivity • Typically outer city ring areas or commuter towns • Detached housing, local shops, fewer intersections	Most of our cycling paths should be classified as outer urban (residential), except those in central area (inner urban) Default: Suburban Geospace: LTA Parking standard Zone 1 comprises of city (restricted zone & Marina Bay) → Urban Geospace: URA Amendment to Master Plan 2019 Land Use layer - Agriculture, Cemetery, Open space, Water Body, Special use → Rural - Business 1, Business 2, Transport Facilities → Industrial - Beach area, park, sports & recreation → recreational

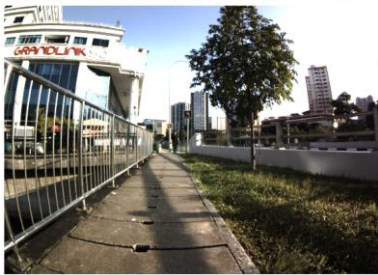
Attribute	Code	Coding guidance	Local guidance
		Rural - Predominantly agricultural or undeveloped land - Sparse or linear development - Few intersections, long distances between buildings - OSM <code>landuse=farmland, forest, grass</code> Industrial - Warehouses, factories, logistics centres - Zoning or land use as <code>landuse=industrial</code> - Often co-located with heavy vehicle routes Recreational - Parks, trails, waterfronts, sporting facilities <code>landuse=recreation_ground, leisure=park, natural=*</code> Common for off-road paths, shared-use facilities away from roads	 Suburban takes precedence if there is an overlap <i>To include an offset of 10m (to improve accuracy); CPN is usually about 1.5m away from the boundary line of land use layer; at development accesses, might not be able to capture, hence to do manually.</i>  [GIS]

Attribute	Code	Coding guidance	Local guidance
Facility type Sidewalk	1	An off-street path intended for pedestrian use only	Footpath 
Multi-use path	2	An off-street path which is intended for use by bicycles/light vehicles and pedestrians. Multiuse paths are generally wider with signage indicating it as such	Shared 2.5m path 
Off-road bicycle path	3	An off-street path which is intended for use by bicycles/light vehicles only	Most of our off-road cycling paths 

Attribute	Code	Coding guidance	Local guidance
On-road bicycle lane	4	An on-street lane intended for use by bicycles/light vehicles, which is out of the main path of large, motorized vehicles and marked accordingly	 Two solid red lines used in city area Eg: TMCR bicycle lane, demarcated bicycle crossing 
Road shoulder	5	Space on the street (but out of the direct path of large, motorized vehicles) used by bicyclists/light vehicles. May be unmarked	Eg: West Camp Road cycling lane, cyclists sharing bus lane; development access, zebra crossing, signalized crossing  
Mixed traffic road lane	6	Bicycles/light vehicles use the same street space/path as large, motorized vehicles (only this category has a risk factor)	 Default: Sidewalk [CV]

Attribute	Code	Coding guidance	Local guidance
Facility access Adequate	1	The facility is accessible and fit for purpose	Bicycle wheeling ramp on POB is coded as adequate, unless you observe that there are cyclists crossing the road and not using the wheeling ramp. Chicanes is coded as adequate as cyclists still can go through (code fixed obstacles present). Construction water filled barricade affect clear width but still passable, code as adequate 
Inadequate	2	Due to congestion, obstructions or other issues, some or all bicyclists and light vehicle users are forced to use alternative route (adjacent paths or roadway). E.g. very high ped flows, very poor surface quality, bollards/fencing which reduce path width for cargo bicycles, lack of ramps. <i>Inadequate affects VB risk.</i> Flag “inadequate” if congestion or blockages detected from: - Obstruction reports - Counterflow or avoidance behaviour Vehicle/bike/pedestrian volume overload	Hoarding by construction or path too crowded (till it forces cyclists to go onto the road) is coded as inadequate; no bypass path behind bus stop forces cyclist to go in front of bus stop or on the road 

Attribute	Code	Coding guidance	Local guidance
<i>Loose or slippery surface</i> Present	1	Record if there is loose gravel, metal plates, wet leaves, a crossfall or any other characteristic present which could result in a bicyclist or light vehicle to lose traction Surface = gravel, leaves, metal plates, etc. occupying more than 1m ² of a 10m coding segment (this may be several smaller patches which amount to more than 1m ²)	There could be some stretches with soil erosion from the adjacent planter box/area Fallen leaves if a lot coded as present 
Not present	2		GIS using Path defects (Algae) [CV still not ready yet] To wait for grip tester results from PATH
<i>Tram or train rails</i> Present	1	Record if tram rails are present on or in the vicinity of the facility which may be crossed by a bicyclist or light vehicle	There is no such path/road in Singapore → 2 (Default)
Not present	2		

Attribute	Code	Coding guidance	Local guidance
Major surface deformation or drain opening Present	1	Record if there is any major surface deformation which could result in a bicyclist or light vehicle to lose balance. Examples include potholes, tree roots, major cracking, drain openings or tarmac edges or ramps with a vertical edge of >3cm	    Uneven drain covers – depends if it will cause a severe or fatal crash.
Not present	2		GIS using PATH's defects layer: tile cracks, tactile tile cracks, uneven pavement

Attribute	Code	Coding guidance	Local guidance
<i>Fixed obstacle on facility</i> Present	1	Record if there is one or more fixed obstacles on the facility that is not movable. Examples include poles, fences, bollards, street furniture, electrical boxes, bus stops, signages, bike racks, shelters, etc	Record if covered linkway poles/OG boxes/ lamp post/VIG posts are within the path width, chicanes. Outgrown vegetation that encroaches onto path.      
Not present	2		[CV]

Attribute	Code	Coding guidance	Local guidance
Non-fixed obstacle on facility Present	1	Record if there is one or more non-fixed obstacle/s on the facility. Examples include rubbish bins, parked cars or bicycles, roadwork signs or device, or other temporary items	Record if there is any non-fixed devices, rubbish bins, motorcycles, advertisement panels, etc 
Not present	2		[CV]
Line of sight Adequate	1	Record if there are any visual obstructions on or around the path which would prevent a cyclist from seeing (or being seen by) an approaching vehicle on a driveway or side road, or another cyclist or pedestrian. Visual obstructions may include shrubs (such as hedges), fences, walls etc. Check visibility from eye height (1.2-1.5m) along the path and toward any driveway, side road, bend or crossing point for • Objects blocking the view withing approximately 10-20m of the conflict point • Continuous objects above 0,6m-0.8m height (e.g. hedges, fences, walls, parked vehicles, dense vegetation) within 1m of the path edge	
Inadequate	2		 Footpath: 12m; Cycling path/shared path: 23m

Attribute	Code	Coding guidance	Local guidance
Delineation Present Not present	1 2	Record if clear edge and central marking (for two-way facilities) are present on the facility. Record as 'not present' if they are absent, only partially present or are in poor condition. Delineation for on-road facilities should only be recorded if provided for a bike lane or road shoulder. Record as 'not present' for mixed traffic road lanes (even if delineation is provided for general traffic)	For most of our newly paved cycling/ shared path, for zebra crossings, signalized crossing Record if cycling/ shared path markings have faded (both sides faded that it cannot be recognized as shared path). Record if it is development access (no delineation). For West camp road (shared bus and cycle lane), code as not present.  [CV first, then if PATH side pass us faded markings, then overwrite]

Attribute	Code	Coding guidance	Local guidance
Light segregation Present	1	Record if there is light segregation between the facility and the road lanes like bollards, flexible posts, or other types of physical object (e.g. a curb) that prevents discretionary vehicle incursion onto a bicycle facility	All our off road paths should have kerb
Not present	2	Record as 'not present' if they are absent, easily traversable (unlikely to discourage/ designed to allow vehicles to cross) or if there are large gaps vehicles can pass through	TMCR on road cycling lane, West camp road shared bus-bicycle lane, zebra crossing, signalized pedestrian crossing, development access, unsignalized junction  [CV/rules]

Attribute	Code	Coding guidance	Local guidance
Facility width per direction Very Narrow less than 1m per direction (<2m)	1	Record the width of the facility for each direction of travel. For 2-way facilities without a centreline, divide the total width by 2 Where a centreline is present, but off-centre (i.e 1 direction has more space than the other) record the narrower of the 2	Footpath 1.5m or 1.8m width [if facility type = footpath + LIDAR FP width <2.0m] If facility type = off road bicycle path + adj sidewalk (0-1m) = absent + LIDAR CYP width <2.0m]
Narrow 1 to 2 m per direction (2-4m)	2		Record for all our 2m CYP+1.5m FP, shared paths, 3.6m shared covered linkway [if facility type = off road bicycle path + adj footpath (0-1m)= present + LIDAR CYP width >=2.0m] [if facility type = Multi use path + LIDAR shared path width <=4m]
Wide more than 2m per direction (>4m)	3		Record for widened 5m shared signalized and zebra ped crossing, 6m pedestrian cum bicycle crossing Display the width of footpath or shared/cycling path (if present) [GIS/Rule]

Attribute	Code	Coding guidance	Local guidance
Adjacent road lane 0-1m Present	1	Record if there is a road lane with live traffic 0-1m from either side of the facility with no protective barrier present	Record if edge of cycling/ shared path is right next to road kerb $\leq 1\text{m}$;  At signalized crossing/development access, this refer to the parallel road 
Not present	2		Park connectors that are away from the road, but along canals, in between parks, etc  [CV+rule]

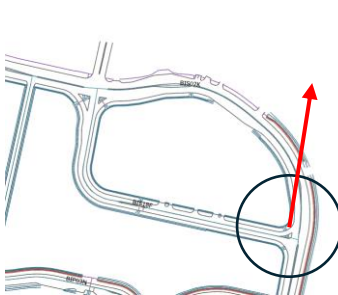
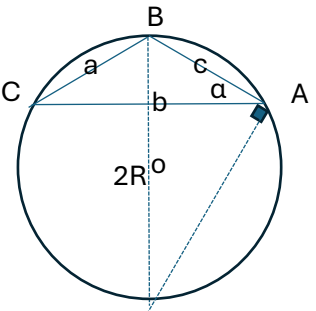
Attribute	Code	Coding guidance	Local guidance
Adjacent vehicle parking 0-1m Present	1	Record if there is vehicle parking 0-1m from either side of the facility with no protective barrier present	Datamall, HDB (point) and URA (polygon) car park lots → now using URA car park lots GDM/ lanemarking, TYP_CD = E (White-dash-centre line on 2way carriageway) [include double yellow line (no parking) to include illegal parking in private landed area]
Not present	2		[GIS]
Adjacent severe hazard 0-1m Present	1	Record if there is a severe hazard 0-1m from either side of the facility. Severe hazards include anything which could cause serious injury or death to a bicyclist or light vehicle user and where there is no protective barrier present, such as waterways or deep, open drains or cliffs or significant drop of >60cm	
Not present	2		[Rare cases/Human intervention]
Adjacent object or level change 0-1m Present	1	Record where there is an object or level change ≤ 60cm high (such as curb) 0-1m from either side of the facility with no protective barrier present	

Attribute	Code	Coding guidance	Local guidance
Not present	2		If path is right beside road carriageway without railing [CV/Rule]
Adjacent sidewalk 0-1m Present	1	Record if there is a pedestrian facility 0-1m from either side of the facility with no physical barrier present (e.g. fence, verge)	Record 1 for footpath side by side dedicated cycling path, pedestrian cum bicycle crossing 
Not present	2		Record 2 for signalized widened PC, shared path Record 2 for split path with > 1m of green verge in between [CV/Rule]
Adjacent road lane 1-3m Present	1	Record if there is a road lane with live traffic 1-3m from either side of the facility with no protective barrier present. Adjacent highways or rural roads with speeds of 80km/h or more (without protective barriers) within 10m of the facility may be recorded	

Attribute	Code	Coding guidance	Local guidance
Not present	2		If green verge is 5m, record as 2 
Adjacent vehicle parking 1-3m Present	1	Record if there is vehicle parking 1-3m from either side of the facility with no protective barrier present	
Not present	2		
Adjacent severe hazard 1-3m Present	1	Record if there is a severe hazard 1-3m from either side of the facility. Severe hazards include anything which could cause serious injury or death to a bicyclist or light vehicle user and where there is no protective barrier present, such as waterways or deep, open drains, or cliffs or significant heights.	
Not present	2		

Attribute	Code	Coding guidance	Local guidance
Adjacent object or level change 1-3m Present	1	Record where there is an object or level change <+ 60cm high (such as curb) 1-3m from either side of the facility with no protective barrier present	Trees
Not present	2		Railing
Adjacent sidewalk 1-3m Present	1	Record if there is a pedestrian facility 1-3m from either side of the facility with no protective barrier present	For split paths
Not present	2		
Grade <5 degrees >=5 degrees	1 2	Record if the facility is on a longitudinal slope of more or less than/equal to 5 degrees	1:25 (4%, 2 degrees) 1:20 (5%, 2.9 degrees) 1:15 (6.7%, 3.8 degrees) 1:12 (8.3%, 4.7 degrees) Get LIDAR 3D point cloud layer from Survey & Land.

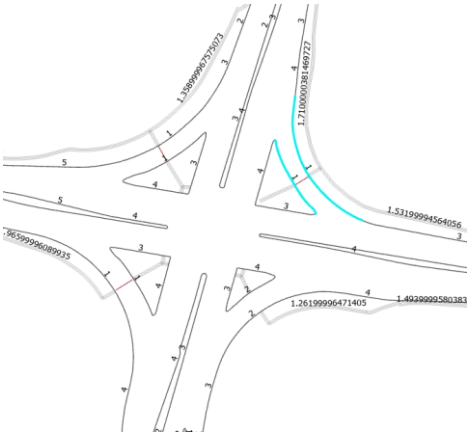
Attribute	Code	Coding guidance	Local guidance
			1. Verify LiDAR points, look for any gaps or corrupted files. 2. Through Python (coding), remove all points that don't have an elevation value 3. Remove any points below -5m and above +170m. This is almost always trees and other non-useful data. 4. Pinpoint exact image data using GPS coordinates. 5. Look for the 50 nearest LiDAR points and identify the lowest elevation one. 6. Remove anything over 1.5m above the lowest point. 7. Find the median of these nearest points that remain below 1.5m of the lowest point. 8. These last few steps above remove any cars, trees, buildings, etc., using median, as over 50% of all points will sit at ground level.

Attribute	Code	Coding guidance	Local guidance
Curvature Sharp turn present	1	Record if the facility has a sharp turn present which would require a bicyclist or light vehicle to slow down significantly to negotiate the corner safely <10m or tapering angle >45 degrees. The 10m radius is from the CROW guide, which states that anything below requires significant slowing down to navigate safely (and that below 5m, it is hard to stay upright).	   Step 1: Create 10m buffer of the image (due to GPS inaccuracy) Step 2: Select LEAST path centreline that is within 10m buffer Step 3: Offset 1m left and right of point to calculate taking area of triangle and hence method Step 4: repeat step 3 with 1m interval for the whole radius of 10m to find the smallest R Area of triangle = $\frac{1}{2} bc \sin \alpha$ Heron method, Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{1}{2}(a+b+c)$ $\frac{\alpha}{\sin \alpha} = 2R \sin 90^\circ = 2R$ $R = \frac{\alpha}{2 \sin \alpha}$ Need to manually sensor out the U-turning movement of the data collector and also cross paths. Buffer reduced further to 5m (to minimize inclusion of opposite paths) Sharp turn or path junction
No sharp turn present	2		[GIS/rules] Footpath (>=6.5m) Cycling path (>=18m)

Attribute	Code	Coding guidance	Local guidance
Street lighting Present Not present	1 2	Record if the facility has adequate lighting at night	Default yes
Pedestrian crossing Present Not present	1 2	Record if there is a designated pedestrian crossing or other feature (such as a bus stop) which necessitates pedestrians crossing the bicycle facility	Record for pedestrian gates or possible pedestrian access that interacts with shared path [manual] For footpath/CPN/shared path, where there is a bus stop [GIS] and perpendicular walkway connecting to the bus stop Also for MRT station exit to bus stops, taxi stands/PUDO/crossings [GIS] Road crossing - Signalised Junction/midblock PC/zebra crossing [GIS/Manual]
Intersecting bicycle facility Present Not present	1 2	Record if there is an intersecting bicycle facility resulting in cross-flow or more bicycle or light vehicle flow joining the facility	Record for intersecting cycling facility [GIS/Manual] Using GIS pedestrian crossing (zebra crossing, signaled pedestrian crossing, bicycle crossing) layer (5m radius)

Attribute	Code	Coding guidance	Local guidance
Intersection approach Shared	1	Record if the approach to a road or street intersection is on-street (shared) 1 or remains separate 2 from traffic until the intersection itself. Eg of shared approaches include facilities ending before the intersection, or where there are pocket lanes for bicyclists	
Separate/NA	2	Record as NA if no road intersection is present	Record 2 for the off road cycling path nearing intersection Default: separate/NA
Intersection or road crossing Present	1	Record if there is a road or street intersection or if the facility crosses a road or street	Zebra crossing, signalized PC, development accesses, unsignalized junctions
Not present	2		[CV]

Attribute	Code	Coding guidance	Local guidance
Crossing facility Present/NA	1	Record if there is a safe crossing facility for bicyclists or light vehicles at the intersection or road crossing. Facilities may include signals, pedestrian crossings, raised crossings, refuge islands etc	Zebra crossing, bicycle crossing, Silver zone kerb cut ramp, two-staged crossing, eyeland considered present as there is traffic calming measure
Not present	2		Record 2 at development accesses unless development puts in formal crossing, refuge island or kerb cut ramps with traffic calming [CV]

Attribute	Code	Coding guidance	Local guidance
Number of lanes – adjacent road 1 per direction/NA >1 per direction	1 2	Record how many lanes are on the road adjacent to the facility. If there is no road, record as NA	LIDAR layer: Kerblines Match the nearest Kerblines (lanes) with the photo, Number of lanes – adjacent road (1 or 2)  [GIS]
Number of lanes – intersecting road 1 per direction/ NA >1 per direction	1 2	Record how many lanes are on the intersecting road. If there is no road intersection, record as NA	LIDAR layer: Kerblines If Facility Type = Mixed traffic road lane, then match the nearest Kerblines (lanes) with the photo, Number of lanes – Intersecting (1 or 2) Else, Number of lanes – intersecting road (1)/NA [GIS]

Attribute	Code	Coding guidance	Local guidance
<i>Property access</i>			
<i>Present</i>	1	Record if there are property entrances or driveways which intersect with the facility	
<i>Not present</i>	2		[pending CV/GIS]

Attribute	Code	Coding guidance	Local guidance
Peak pedestrian flow along or across facility		<i>Flows and speed data are critical to the model. We could simulate scenarios and identify volume thresholds for risk, but getting data from official sources or collecting samples is always recommended. Pedestrians and bicyclists need to be collected in peak hours on a typical movement day, while vehicle volumes are on a daily basis.</i>	
None	1	If recorded as 'None', no Bicycle-pedestrian crash type score will be produced)	
Low	2	Free flow speed can be maintained during peak periods.	Zebra crossing/signalised PC [auto] Perpendicular pedestrian crossing, (CV/rules), bus stops (GDM layer), can also use LOOK marking as a gauge as low 
Moderate to high	3	Free bicyclist flow is disrupted during peak periods.	High – along schools, hawker centres/town centres/MRT (GDM MRT exit) - 2025/bus interchanges For MRT exit points, need to offset 20-25m front and back to include nearby stretches [GIS]

Attribute	Code	Coding guidance	Local guidance
Peak bicycle/ LV traffic flow <i>Low</i> <i>Moderate to high</i>	1 2	Record bicycle/light vehicle flows during peak times, when most activity is present (e.g. morning or evening peak hours or school finish times) Steady or busy flows during peak times Thresholds can align with >50 bikes/hr or >100 users/day as indicative	AMG's 24/7 sensors (10) with rotation + before and after counts [GIS] Cyclist, (LV) PMD, PAB; PMA is considered as pedestrians
Observed proportion of cargo bikes and mopeds <i>Low</i> <i>Moderate to high</i>	1 2	Record the proportion of Type B (cargo bike) or D (25-35kph + >35kg) vehicles using the facility	D is equivalent to motorcycles as all our ePMD, PAB < 20kg Default: low

Attribute	Code	Coding guidance	Local guidance
Bicycle / LV speed – average <20km/h >= 20km/h	1 2	Record the average free flow speed of bicycles and light vehicles	Record 1 for normal shared/CPN Record 2 for long, uninterrupted PCN, cycling bridge/underpass (with wide path) Default 1 Future A webpage app developed by AI office, has added object speed detection function (besides the traffic count function), can run speed detection from soe of the sensor sample videos if needed; currently the 'layer' is the tableau dashboard, haven't convert into a shape file or GIS layer yet
Bicycle / LV speed differential <10km/h >=10km/h	1 2	Record where the standard deviation of the free flow speed of bicycles and light vehicles is greater or less than 10km/h	Default 1

Attribute	Code	Coding guidance	Local guidance																																								
Road AADT Number	NA	Record the average annual daily traffic (AADT) of the road (if present). If a road is not present, record it as zero Figure 5 CycleRAP AADT and speed risk multipliers <table> <tr> <th>AADT</th><th>Multiplier</th><th>AADT</th><th>Multiplier</th></tr> <tr> <td>0</td><td>0.250</td><td>15000</td><td>1.403</td></tr> <tr> <td>100</td><td>0.250</td><td>17500</td><td>1.501</td></tr> <tr> <td>500</td><td>0.500</td><td>20000</td><td>1.606</td></tr> <tr> <td>1500</td><td>0.750</td><td>22500</td><td>1.718</td></tr> <tr> <td>2500</td><td>1.000</td><td>25000</td><td>1.838</td></tr> <tr> <td>5000</td><td>1.070</td><td>30000</td><td>1.967</td></tr> <tr> <td>7500</td><td>1.145</td><td>35000</td><td>2.105</td></tr> <tr> <td>10000</td><td>1.225</td><td>40000</td><td>2.252</td></tr> <tr> <td>12500</td><td>1.311</td><td></td><td></td></tr> </table>	AADT	Multiplier	AADT	Multiplier	0	0.250	15000	1.403	100	0.250	17500	1.501	500	0.500	20000	1.606	1500	0.750	22500	1.718	2500	1.000	25000	1.838	5000	1.070	30000	1.967	7500	1.145	35000	2.105	10000	1.225	40000	2.252	12500	1.311			To get from iTransport by stretches and reference Mytransport/Traffic news to infer Gotten from Datamall (traffic flow – hourly average traffic flow from a representation month of every quarter during 0700-0900h) → use Postman to access [However, the data is not islandwide, it is based on available detectors] To get from ERP2 data (7am-9am, any typical weekday) Peak hour *8*350/365= AADT (PHF is from planning division) 2500 (Cat 5 private landed roads) 5000 9000 15000 [If the path is segregated by an underpass then the expressway, exclude the AADT of the expressway] [pending ERP2 data]
AADT	Multiplier	AADT	Multiplier																																								
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Attribute	Code	Coding guidance	Local guidance
Heavy vehicle flow <i>Low/restricted access</i>	1	Record the level of heavy vehicle flow present on the road. If a road is not present, record it as 'low/restricted access' Streets where heavy vehicles are restricted all or part of the time, may include local traffic zones or specific restrictions. If restrictions are part time, they should correlate with daylight hours and evenings when bicycles and light vehicles are in highest use; Local, neighbourhood streets where there is very low commercial traffic or no bus routes; Country roads or lanes with no heavy vehicle use	
<i>Moderate to high</i>	2	A bus route is present along the road; the road is an industrial or commercial area (without restrictions); the road is a major corridor (e.g arterial road or trunk road) or highway (without restriction)	GDM: lane marking layer – markings for bus lanes: A1, A6, A7, L, Q, Q1, Q2, Y (10 Jun 25) [GIS]
Road speed limit number	NA	Record the road speed limit. If no road is present, record as 'NA' (Speed limit does not affect risk scores. It is used for analysis purpose only.)	From GDM layer [2015] [GIS]

Attribute	Code	Coding guidance	Local guidance
Road operating speed (mean) number	NA	Record the mean operating speed of motorized traffic on the road. If no road is present, record as 'NA' Average speed at which motorized vehicles travel in free flow conditions along the road. Mean speed may be smoothed along homogenous lengths of road, but should reflect the changes in operating speeds due to traffic calming measures, intersection design and other design features	To get average speed for some major arterial roads from iTransport. Then use an off peak MyTransport app/Traffic news red-amber-green heatmap as a proxy to scale the speeds for other roads From Datamall: Traffic Speed Bands → use Postman to access [Data is not complete, not islandwide] From ERP2 data → Any weekday 10-11am [Need to cater for OBU adoption rate] For 50kph roads, midblock: 45kph, at development access & 1 frame after sign. intersection: 40kph, slightly before and after hump, signalized intersection: 30kph If speed <1k/h, there appears to be some errors. [GIS]

Grouping appearance of attributes

<u>Facility configuration</u> Area type Facility type Adjacent sidewalk 0-1m Adjacent sidewalk 1-3m Adjacent road lane 0-1m Adjacent road lane 1-3m Adjacent vehicle parking 0-1m Adjacent vehicle parking 1-3m Adjacent object or level change 0-1m Adjacent object or level change 1-3m	<u>Facility clear width</u> Facility Access Fixed obstacle on facility Non-fixed obstacle on facility Facility width per direction Width restrictions Light segregation Adjacent severe hazard 0-1m Adjacent severe hazard 1-3m Line of sight	<u>Facility surface conditions</u> Delineation Major surface road deformation Loose or slippery surface Grade Curvature Tram or train rails Street lighting
<u>Intersection</u> Intersection approach Intersection or road crossing Crossing facility Property access Pedestrian crossing Intersecting bicycle facility Number of lanes – adjacent road Number of lanes – intersecting road	<u>Flow & Speed</u> Flow direction Peak pedestrian flow along or across Peak bicycle/LV traffic flow Obs proportion of cargo bikes Bicycle/LV speed average Bicycle/LV speed differential Road AADT Road Operating speed (mean)	

GIS Shp file layers used in tool

- MRT_EXITS.shp - Peak pedestrian flow along, across facility, pedestrian crossing
- Bus lanes.shp - Heavy vehicle flow
- URA_PARKING_LOT.shp - Adjacent Vehicle Parking 0-1m
- LanduseRural2025.shp (rural) - width and curvature
- LanduseIndustrial2025.shp (industrial) - width and curvature
- CentralMB2025.shp (inner_urban) - width and curvature
- ROADATTRIBUTELINE_SPEEDLIMITS.shp - road speed limit
- 31Oct24_Link_FUL.shp - Road Operating Speed
- Bus_Stop.shp – pedestrian crossing
- Road_crossing.shp – Pedestrian crossing

1) After computing the sheet, it would be good to relook at the coding again to adjust for relative comparison across segments or to clean up incorrect entry in the Upload_data sheet directly

2) Calculate the CycleRAP score

VB crashes:

- Low risk: <10 (displayed in green)
- Medium risk: 10 to 25 (displayed in yellow)
- High risk: 25 to 60 (displayed in red), and
- Extreme risk: > 60 (displayed in dark red).

Other types of crashes (BB, BP, SB)

- Low risk: <5 (displayed in green)
- Medium risk: 5 to 10 (displayed in yellow)
- High risk: 10 to 20 (displayed in red), and
- Extreme risk: >20 (displayed in dark red).

The CycleRAP score (i.e. the sum of the crash types) is displayed as being low, medium, high or extreme risk only where one or more of the crash types fall into that category. That is, the CycleRAP score will only be classified as 'Medium risk' if one or more of the crash types present medium risk scores, and 'High risk' if one or more of the crash types present high risk scores.